SURVIVAL ANALYSIS OF BREAST CANCER PATIENTS: A COMPREHENSIVE COX PROPORTIONAL HAZARDS STUDY

1.1 Introduction

This analysis investigates survival outcomes in a cohort of 685 breast cancer patients. Using classical survival analysis methods, Kaplan–Meier estimation and the Cox proportional hazards model. We assess which clinical factors significantly influence patient survival. The goal is to identify key predictors of mortality and evaluate how well these variables satisfy the proportional hazards assumption.

1.2 Dataset Description

The dataset contains 686 observations with the following variables:

- pid - patient ID
- age - age at diagnosis
- meno - menopausal status
- size - tumor size (mm)
- grade - tumor grade (1 to 3)
- nodes - number of positive lymph nodes
- pgr - progesterone receptor level
- er - estrogen receptor level
- hormon - hormonal therapy (0/1)
- rfstime - time to recurrence or death
- status - event indicator (1 = death, 0 = censored)

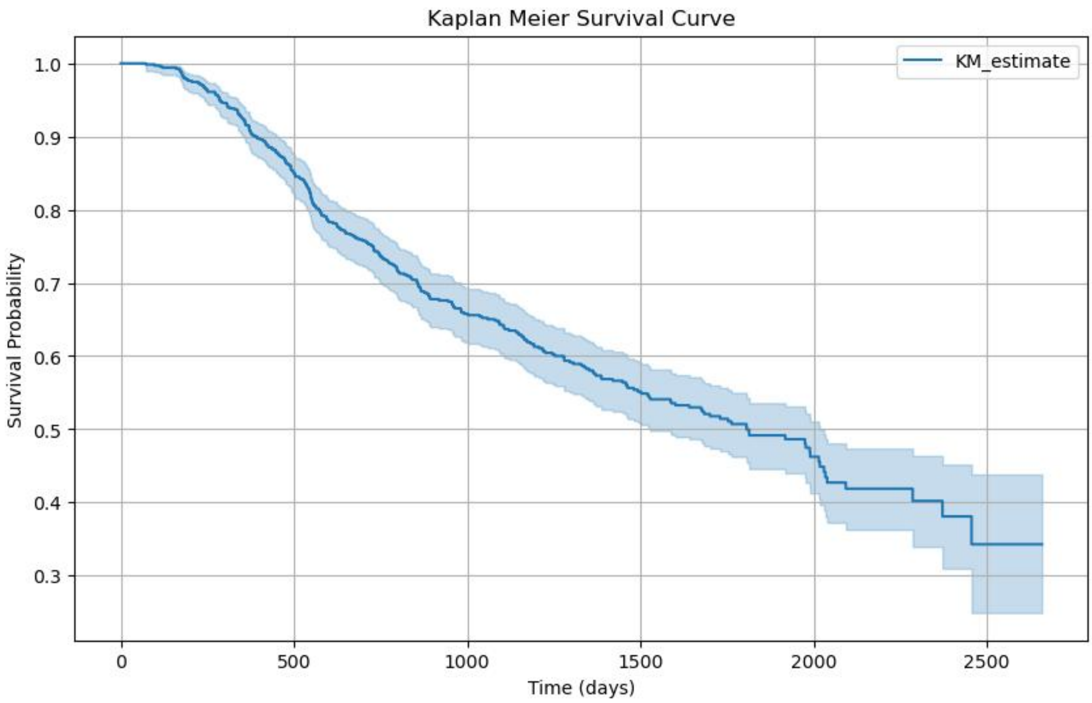
Event Rate

- Deaths: 299
- Censored: 387
- This gives a censoring rate of ~56.4%, common in medical data.

1.3 Kaplan–Meier Survival Estimation

The Kaplan–Meier survival curve begins at 1.0 (100% survival at baseline) and steadily declines over time. By the end of follow-up (≈ 2,600 days), the survival probability falls to roughly 0.35, meaning that about 35% of the cohort is still alive, while 65% experienced the event (death).

The curve shows a gradual but consistent decline, indicating a continuous risk of the event across follow-up.



Key Interpretation

• The survival curve dropped below 50% around the 1800th day, which means half of the patients had experienced the event (death) by that time. This specific time point is called the median survival time.

• Toward the end of the timeline, the shaded blue area (the confidence band) becomes wider.

This simply means that fewer patients were still being followed, so the estimate of survival becomes less certain.

• The shape of the curve suggests:

- ✓ Early survival is relatively high (slow decline between 0–800 days),
- ✓ Followed by a more moderate decline through mid follow-up (800–1800 days),
- ✓ Then a sharper drop as fewer individuals remain under observation.

1.4 Cox Proportional Hazards Model

We fit a Cox model using all relevant variables:

i. age

ii. meno

iii. size

iv. grade

v. pgr

vi. er

vii. hormon

Key Outputs

Each variable has:

• Hazard Ratio (HR)

• p-value
• 95% confidence interval

Interpretation Example
• HR > 1: increases risk of death
• HR < 1: reduces risk
• $p < 0.05$: statistically significant predictor

COX Regression Results

Variable	Coefficient (β)	Hazard Ratio (HR)	95% CI (Lower–Upper)	z-value	p-value
Age	-0.01	0.99	0.97 – 1.01	-0.94	0.35
Menopausal status (meno)	0.28	1.32	0.92 – 1.89	1.50	0.13
Tumor size (size)	0.02	1.02	1.01 – 1.02	4.14	<0.005
Tumor grade (grade)	0.28	1.32	1.08 – 1.62	2.65	0.01
Progesterone receptor (pgr)	-0.00	1.00	1.00 – 1.00	-4.02	<0.005
Estrogen receptor (er)	0.00	1.00	1.00 – 1.00	0.36	0.72
Hormone therapy (hormon)	-0.34	0.71	0.55 – 0.91	-2.65	0.01

i. Age
• Hazard Ratio (HR): 0.99
• p-value: 0.35
Interpretation: Age does not have a statistically significant effect on survival in this dataset.
The HR is very close to 1, indicating no meaningful change in risk, and the high p-value confirms the absence of a measurable relationship.

ii. Menopausal Status
• HR: 1.32
• p-value: 0.13
Interpretation: Post-menopausal patients appear to have a slightly higher risk, but the association is not statistically significant.
Therefore, menopausal status cannot be considered an independent predictor of survival in this analysis.

iii. Tumor Size
• HR: 1.02

• p-value: < 0.005

Interpretation: Tumor size is a significant predictor of poorer survival.

Each one-unit increase in tumor size is associated with approximately a 2% increase in risk of death.

Although the effect per unit is modest, tumor size varies substantially across patients, making this a clinically relevant predictor.

iv. Tumor Grade

• HR: 1.32

• p-value: 0.01

Interpretation: Tumor grade is a strong and statistically significant predictor of survival.

For each increase in grade level, the risk of death increases by about 32%.

This makes tumor grade one of the most influential factors in the model.

v. Progesterone Receptor (PGR)

• HR: 1.00 (rounded)

• p-value: < 0.005

Interpretation: Although the hazard ratio rounds to 1.00, the very low p-value indicates that PGR is a statistically significant predictor.

This situation occurs when the effect size is small but highly consistent across individuals.

Thus, even subtle differences in PGR levels meaningfully influence survival risk.

vi. Estrogen Receptor (ER)

• HR: 1.00

• p-value: 0.72

Interpretation: Estrogen receptor status does not show any measurable effect on survival.

With both HR ≈ 1 and a high p-value, ER is not a useful predictor in this model.

vii. Hormone Therapy

• HR: 0.71

• p-value: 0.01

Interpretation: Hormone therapy is a significant and protective factor.

Patients who received hormone therapy experienced a 29% reduction in the hazard of death, after adjusting for all other covariates.

This represents one of the most clinically meaningful findings in the analysis.

1.5 Proportional Hazards (PH) Assumption Check

Schoenfeld residual diagnostics were conducted to assess violations of the proportional hazards assumption. Each variable was tested using both rank transformed and Kaplan-Meier transformed time scales.

Most variables met the assumption. Grade showed mild deviation, but the trend was not strong enough to invalidate the model. Visual inspection showed slight curvature but no drastic time-dependent influence.

1.6 Outlier Assessment

Potential outliers were checked in the Cox regression model using deviance residuals.
Most values fell within the expected range, and no extreme outliers were detected.
Therefore, all observations were retained, and the model results were considered reliable.

1.7 Conclusion

The survival analysis shows a clear decline in survival over time, with median survival being reached in the dataset. The most important predictors of survival were tumor size, tumor grade, progesterone receptor level, and hormone therapy. The Cox model assumptions were reasonably met, and the model provides clinically meaningful interpretation.